SMART INDIA HACKATHON 2026

TITLE PAGE

Problem Statement ID: SIH26249

Problem Statement Title: Air Power — Predictive Maintenance & Fleet Availability

Organisation: Ministry of Defence

Idea: **FLIGHTLINE** — Fleet Availability Twin

Team Name: Code Byte

Code Byte

IDEA TITLE

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FLIGHTLINE — Fleet Availability Twin

A decision layer that joins the four maintenance silos, predicts how many sortie-days the fleet is about to lose, and names the action that saves the most of them — scored in aircraft-days, not RMSE.

THE PROBLEM :

The statement names fragmentation, not accuracy: *"fragmented and largely reactive … not adequately integrated".*

- Health monitoring, technical records, spares and MRO agencies never meet
- A fault predicted on Tuesday collides with a six-week spares lead time — the aircraft sits on the ground anyway
- RUL models report RMSE; nobody reports aircraft-days

FLIGHTLINE :

ONE GRAPH OF THE FOUR SILOS

tail → LRU → part → stock → depot bay → sorties

PROGNOSIS WITH HONEST INTERVALS

Remaining life as a 90% conformal interval; abstains when unsure

FLEET SIMULATOR + OPTIMISER

90 days, day by day; CP-SAT re-sequences depot inductions and expedites

CANNIBALISATION, PRICED

Every swap returns a net in aircraft-days over 30 days

WHAT THE OFFICER GETS :

- Ranked actions, each scored in aircraft-days gained
- "Why this number?" — the trace of which silo said what
- Availability attribution: why Ao is what it is
- Live what-if: ground aircraft, delay spares, surge sorties
- Printable draft maintenance order

It ranks and prices. It never authorises — the signature stays with the officer.

THE REFRAME: APH

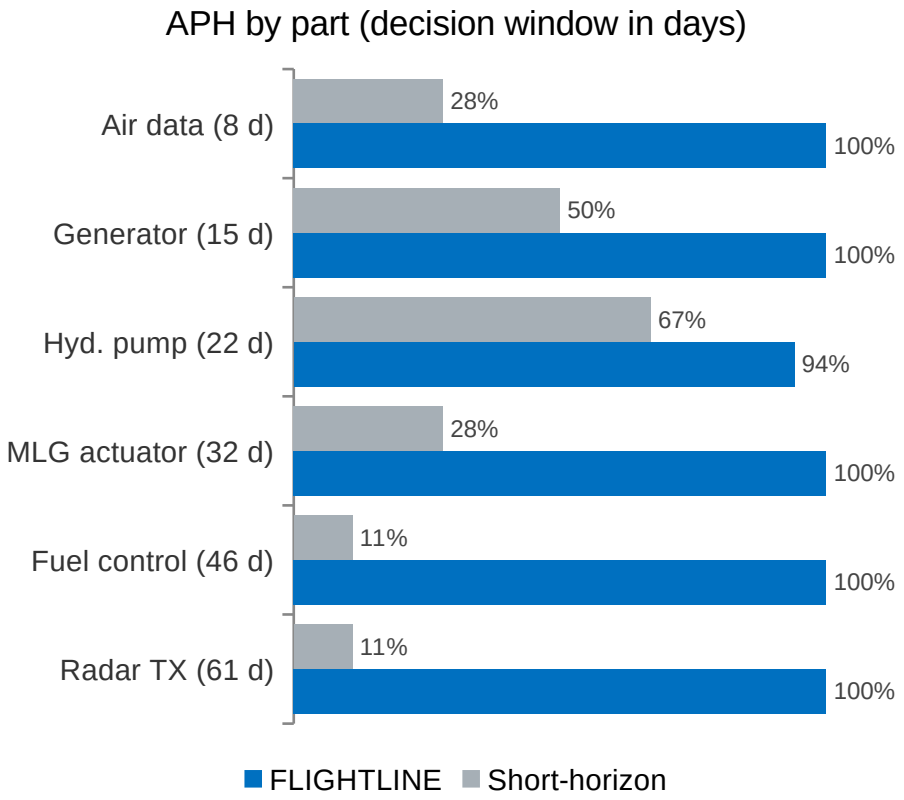
$$APH = P(t_{warning} \leq t_{failure} - (\text{spares lead} + \text{depot queue} + \text{bay time}))$$

A prediction is worth nothing if it arrives after the last day you could have acted on it.

Two real models · same training data · same 108 held-out run-to-failure units

Measure	Short-horizon model (RUL capped at 15, literature convention)	FLIGHTLINE (uncapped + conformal)
RMSE on the model's own target	0.9 cycles	**48.6 cycles**
RMSE in the last 15 cycles before failure	3.7 cycles	**7.1 cycles**
APH — failures warned in time	32%	**99%**
Aircraft-days lost to late warnings / 100 failures	877	**9**
Fleet backtest: sorties lost in 90 days (no prognosis: 18)	12	**8**

More accurate near failure — and still loses. The capped model cannot see past 15 cycles (≈22 days); most spares take longer than that to arrive. FLIGHTLINE warns at a median 1.8× the decision window.



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TECHNICAL APPROACH

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HEALTH MONITORING
144 installed LRUs · multivariate sensor histories, 6 operating regimes (N-CMAPSS shape)

TECHNICAL RECORDS
14 phase inspections due · durations · cannibalisation history

SPARES
18 stock locations · lead times 7–60 days · orders in transit

MRO AGENCIES
2 depots · 3 bays · one induction crew per depot per day

FLEET TWIN one NetworkX graph · 200 nodes · 392 edges · tail → installed LRU → part number → stock location → depot bay → sortie plan

1 PROGNOSIS
- HistGradientBoosting on rolling mean / std / slope per sensor + age
- Mondrian conformalised quantile regression → 90% RUL interval
- Abstains when interval > decision window

2 SIMULATOR
- 90 days, day by day; wear ∝ sorties flown
- Base-stock (S−1, S) spares, FIFO
- Every down-day = one labelled "hold" → exact attribution

3 OPTIMISER
- OR-Tools CP-SAT: induction day + depot, expedites
- Max sortie-days, then aircraft-days
- 2 passes + simulator check; deterministic budget

4 DECISION LAYER
- Cannibalisation priced over 30 days
- Spare unlock ranking (monotone replay)
- Actions in aircraft-days + causal trace

5 DELIVERY
- FastAPI + offline dashboard
- Live what-if, re-solves in ≈1 s
- Printable draft maintenance order

TECHNOLOGY STACK

DATA	ML	OPTIMISATION	GRAPH	SERVING	ASSURANCE
SQLite · seeded generator · N-CMAPSS / C-MAPSS loader	scikit-learn HistGradientBoosting · split-conformal	Google OR-Tools CP-SAT	NetworkX	Python · FastAPI · uvicorn · Chart.js (vendored)	77 pytest tests · same seed → same numbers

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CANNIBALISATION, PRICED

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What GAO-02-86 found (US Navy + Air Force, FY1996–2000)

≈850,000	cannibalisations reported
≈5.3 million	additional maintenance hours
≥ 2×	“at least twice as long as normal repairs”
111 parts	missing from one aircraft
903–1,756 days	four cannibalised FA-18s not flown
> 25%	readiness fell when the Army stopped (1987 test)

Services' own policy: *cannibalisations “aggravate the very same supply problems that they are meant to overcome.”* **Load-bearing and corrosive at once.**

Source: GAO-02-86, Military Aircraft: Services Need Strategies to Reduce Cannibalizations, Nov 2001.

So FLIGHTLINE prices it

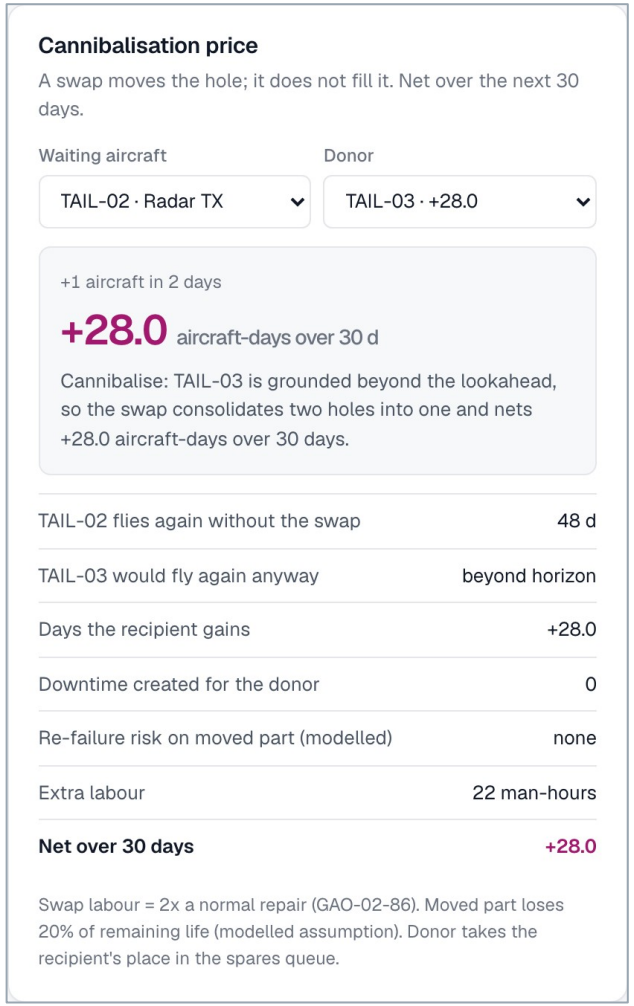
net = recipient gain − donor loss − re-failure

The donor takes the recipient's place in the spares queue: the swap moves the hole, it does not fill it. Swap labour 2× normal (GAO). Moved part loses 20% of remaining life — a modelled assumption, labelled on screen.

Same radar transmitter for grounded TAIL-02:

Donor TAIL-03 grounded for the horizon	**+28.0**
Donor TAIL-06 down 11 more days anyway	**+8.9**
Donor TAIL-03 flying today	**−2.0**

aircraft-days over 30 days



Live card in the prototype (TAIL-01 and TAIL-03 grounded)

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THE ABSTENTION GATE

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We can tell you when we don't know.

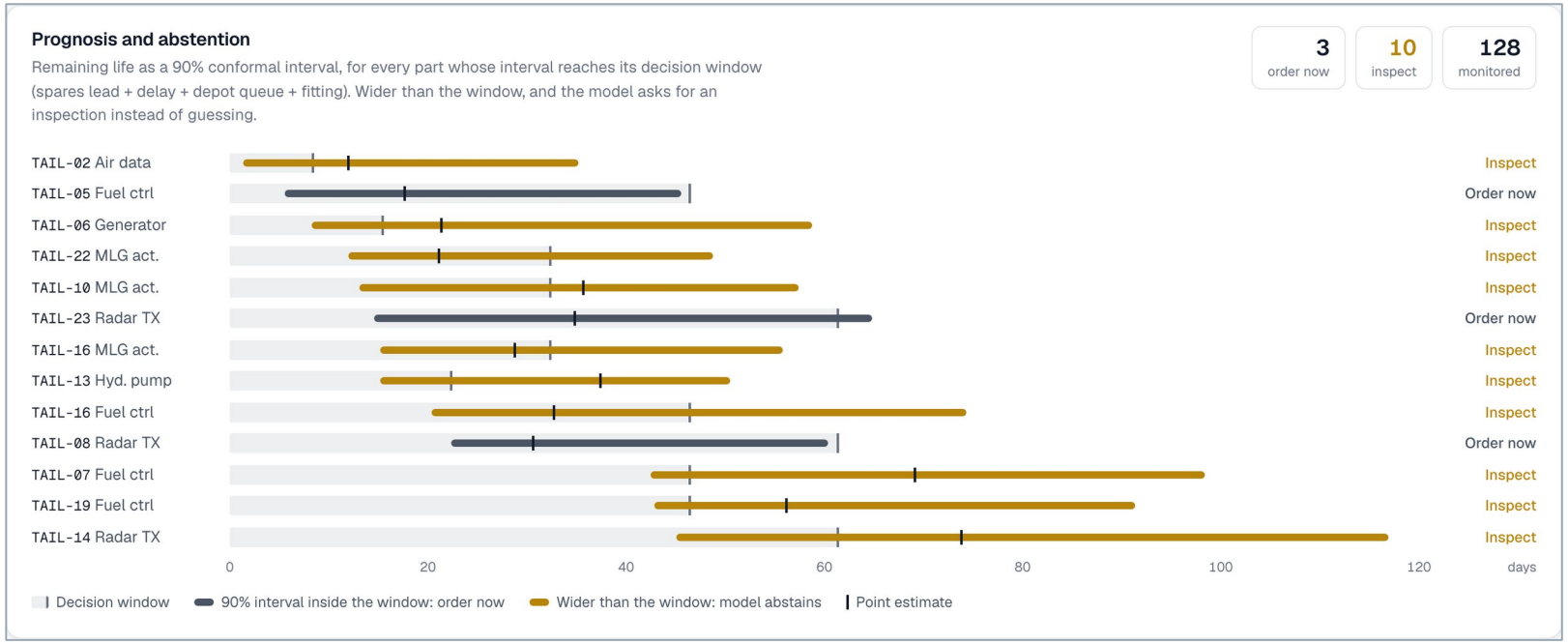
Rule
Abstain when the 90% interval is wider than the decision window **and** its lower edge falls inside it.
→ recommend inspection, not a prediction

91.6% coverage on 1,296 held-out points (target 90%)

94% coverage in the last 20 cycles before failure (85 points)

10 parts flagged for inspection today; 3 confident enough to order; 128 monitored

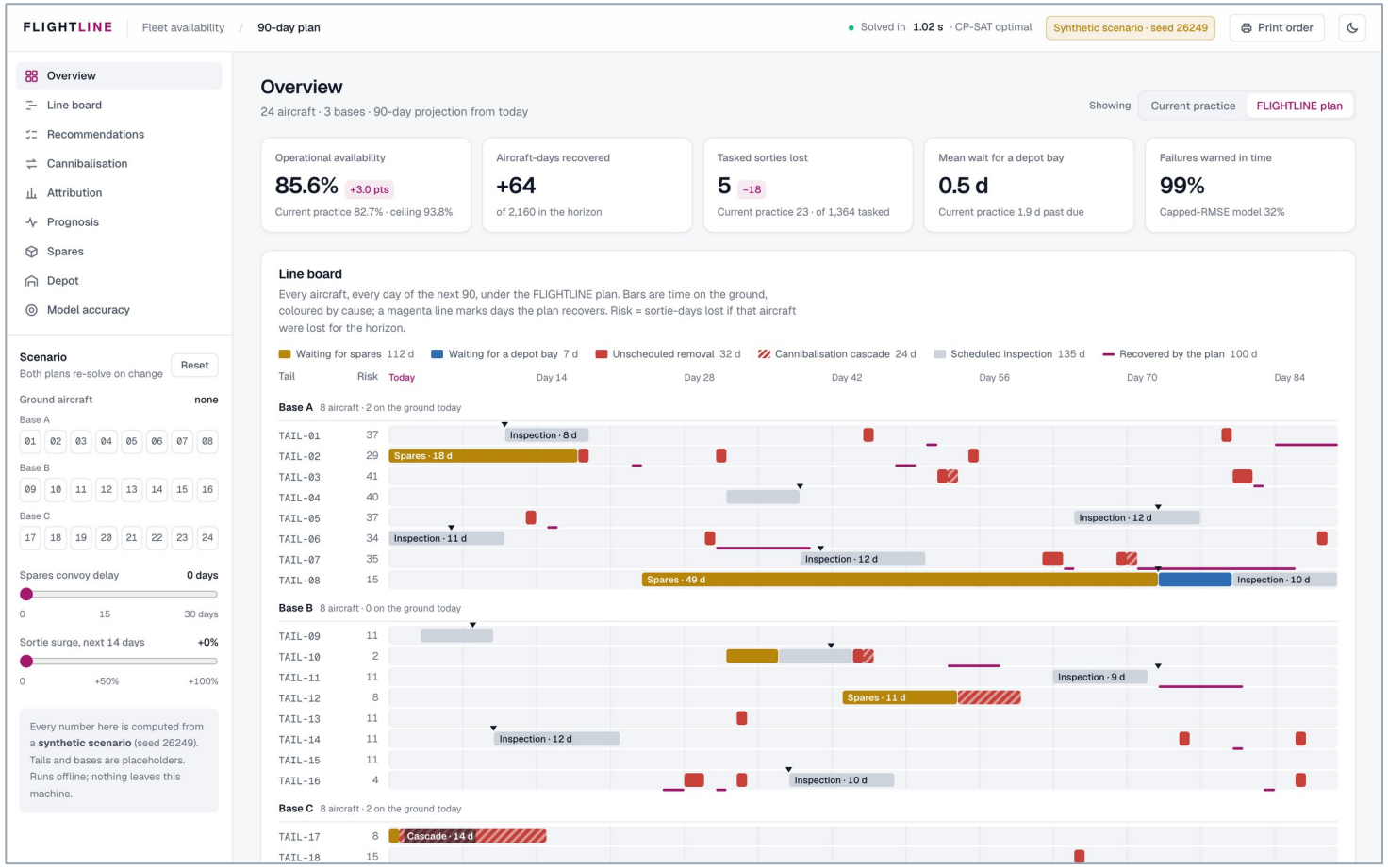
Mondrian conformalised quantile regression: calibrated separately in three bands of predicted RUL, so the guarantee holds near failure, not just on average.



Dark bar = 90% interval narrow enough to act on: order now. **Amber bar** = wider than its decision window (shaded): the model abstains.

Where a false negative can cost aircrew and a false positive wastes a sortie, a system that knows the limits of its own confidence is worth more than one that is slightly more accurate on average.

LIVE DEMO



Running prototype · 127.0.0.1:8000 · offline · seed 26249 · re-solves in ≈1.0 s

Ao 82.7% → 85.6% · +64 aircraft-days · +18 sorties flown · sorties lost 23 → 5

Let the panel break it

“Ground two aircraft at Base A.”
Click 01 and 03

Ao falls to 79.0%. Top action becomes: cannibalise the radar from grounded TAIL-03 to TAIL-02, +28.0 aircraft-days.

“Spares convoy is 10 days late.”
Drag convoy delay to 10

Spares share of the gap 8.0 → 9.9 pts; order-now parts 3 → 10; priced cannibalisation +60.

“Surge: 40% more sorties.”
Drag sortie surge to +40%

Sorties lost: 64 under current practice vs 39 under the plan (+83 aircraft-days).

Every action carries “Why this number?” — the trace of which silo said what.

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FEASIBILITY AND VIABILITY

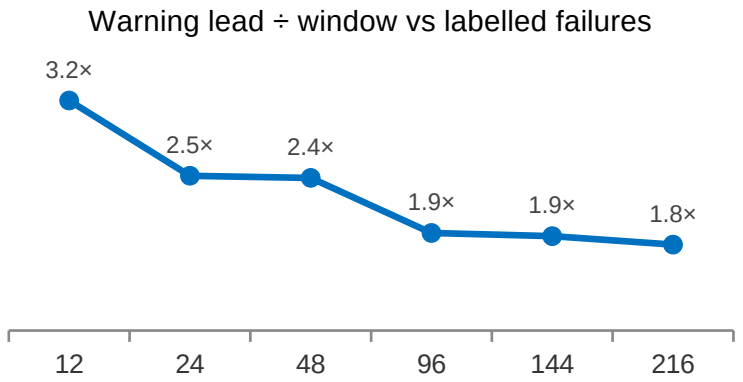
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HONEST ABOUT DATA

- We have no IAF data and do not pretend to. The badge says SYNTHETIC SCENARIO on every screen
- Degradation: 432 run-to-failure units with the shape of NASA N-CMAPSS
- Drop real N-CMAPSS / C-MAPSS files in data/raw and the model retrains on real engines; logistics stay synthetic and the badge says so
- Decisions never see ground truth; it is used only to backtest

PLATFORM-AGNOSTIC

Swap the degradation model; everything above it is unchanged. Fewer labelled failures do not make it unsafe — the interval widens and it warns earlier:



APH stays 95%–100% from 12 to 216 labelled failures. More data buys efficiency, not safety.

RISKS → MITIGATIONS

Model ≠ real fleet	Decision layer is model-agnostic; conformal recalibrates on a small held-out set
Classified data	Air-gapped on one base server; no cloud, no external API
Trust	Ranks and prices, never authorises; every number traceable
Optimiser approximations	Every plan re-scored by the full simulator

RESEARCH AND REFERENCES

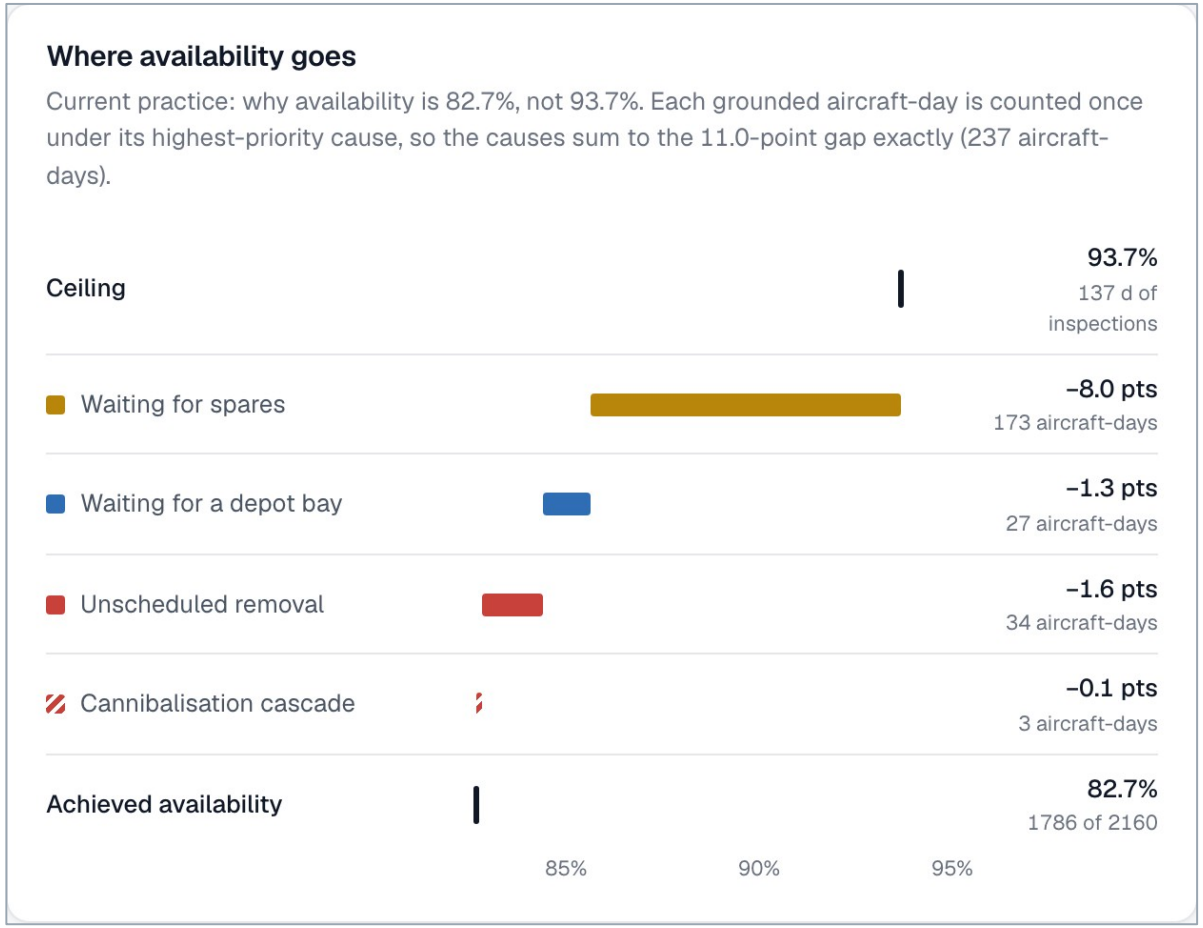
- Arias Chao M. et al., Aircraft engine run-to-failure dataset under real flight conditions (N-CMAPSS), Data 6(1):5, 2021 · Saxena A. et al., C-MAPSS, PHM 2008
- Romano Y., Patterson E., Candès E., Conformalized Quantile Regression, NeurIPS 2019
- US GAO, GAO-02-86, Military Aircraft: Services Need Strategies to Reduce Cannibalizations, 2001 · Sherbrooke C., METRIC, Operations Research 16(1), 1968
- Google OR-Tools CP-SAT solver · scikit-learn HistGradientBoostingRegressor · NetworkX

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AVAILABILITY ATTRIBUTION

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Why is availability 82.7% and not 93.7%?



Cause	Points of Ao	Aircraft-days
Waiting for spares	**8.0**	173
Waiting for a depot bay	**1.3**	27
Unscheduled removal	**1.6**	34
Cannibalisation cascade	**0.1**	3
Total gap below the ceiling	**11.0**	**237**

Every grounded aircraft-day is counted once, under its highest-priority reason, so the causes add up to the gap exactly. It is a unit-tested invariant, not a chart.

This is the screen a Base Maintenance Officer opens at 0700.

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IMPACT AND BENEFITS

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+64

aircraft-days in 90 days on a 24-aircraft fleet (+3.0 pts Ao), same aircraft, parts and bays

23 → 5

tasked sorties lost to unavailability, current practice → plan

1.9 → 0.5

mean days an inspection waits for a depot bay

99%

of failures warned in time to act (capped-RMSE model: 32%)

BASE MAINTENANCE OFFICER

- Why availability is what it is, by cause
- Which depot induction to move, and when

LOGISTICS OFFICER

- Which spare, stocked where, unlocks the most
- Which order to expedite, with its aircraft-days

ENGINEERING OFFICER

- Cannibalisation priced before it is signed
- Inspect-or-order, with the model's own uncertainty

DEPLOYMENT

Air-gapped by construction: one base server, one SQLite file, open-source stack. No cloud call, no LLM, no GPU, no foreign dependency. Nothing leaves the base network.

WHAT WE NEED NEXT

- Labelled removals for one LRU family
- Lead-time history and stock levels by base
- Bay counts and phase intervals for one type; cannibalisation logs